PPP-P-1655A
August 24, 1973
SUPERSEDING
Int. Fed. Spec. PPP-P-001655
August 11, 1971

FEDERAL SPECIFICATION

PAIL, PLASTIC, SHIPPING AND STORAGE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers 5 gallon plastic pails for the shipping and storing of liquids and dry materials.

1.2 Classification. The plastic pails shall be furnished in the following types, classes and grades (see 6.2).

1.2.1 Types and classes. The plastic pails shall be furnished in the following types and classes:

Type I - Open-head
  Class 1 - with steel cover
  Class 2 - with plastic cover
Type II - Tight-head

1.2.2 Grades. The plastic pails shall be furnished in the following grades:

Grade GP - General Purpose except Food and Beverage.
Grade FB - Food and Beverage.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320 - Fiberboard: Sheet, Stock and Cut Shapes.

Federal Standards:

Fed. Test Method Standard No. 101 - Preservation, Packaging and Packing Materials; Test Procedures.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cummulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available with charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods Of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

49 CFR 178.19

Transportation

21 CFR 121
Federal Food, Drug and Cosmetic Act and
Regulations Promulgated Thereunder

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations maybe obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

D 638 (68) - Test for Tensile Properties of Plastics.
D 747 (70) - Test for Stiffness of Plastics by Means of Cantilever Beam.
D 1505 (68) - Test for Density of Plastics by the Density-Gradient Technique.
D 1603 (68) - Test for Carbon Black in Ethylene Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Plastic. The pails shall be formed of high density polyethylene. The polyethylene shall have the following properties:

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Property	Specification	Paragraph
Density range	0.941 - 0.965	4.4.3
Tensile strength	3000 p.s.i. min.	4.4.4
Percent elongation	75% min.	4.4.5
Stiffness	65,000 p.s.i. min.	4.4.6
Carbon black, amount (when specified)	1.0%	4.4.7

3.1.1.1 Ultraviolet light protection. Ultraviolet light protection for the pails shall be provided (by impregnation of the polyethylene with carbon black, or other equally efficient pigments or inhibitors) when specified by the purchaser.

3.1.1.2 Grades. The pails shall be of the grade required for the product(s) intended to be filled.

Grade GP - For general purpose use. A polyethylene compound or a combination of compounds with or without additives or other modifying substances, required to contain the chemical, compound or related product to be filled in the plastic container, that will neither cause nor contribute to an undesirable reaction between the contents and the container.

Grade FB - For food and beverage use. A polyethylene compound or a combination of compounds, with or without additivies or other modifying substances, in complete compliance with the Food and Drug Administration Regulations governing the raw materials permitted for the fibrication of packaging materials used in the packaging of food and beverage. Any compound, combination of compounds, additivies, modifying substances, or equal process intended to change the physical characteristics or capabilities of the plastic containers that is known to cause or contribute any toxic or other harmful effect to the food or beverage contained is prohibited not withstanding that such compound, combination of compounds, additivies, modifying substances or special process may not specifically be prohibited by the Food and Drug Regulations.

3.1.2 Steel wire. The bail (type I pail) shall be galvanized steel wire. Wire handle when used (type II pail) shall be galvanized steel wire.

3.2 Construction.

3.2.1 Body (type I, class 1). The sides and bottom of the pail shall be integral unit not less than 0.090 inches thick. The top rim of the pail shall be curled down and may contain a reinforcing wire. The pail shall be tapered to nest to conserve space in storage and shipment. The interior surfaces of the pail shall be smooth and free of all cracks, crevasses, and recesses wherein product might be retained when pail is emptied. When fitted with covers, pails shall be designed so that the pails can be easily stacked.

3.2.2 Bails (type I, class 1 and class 2). The pails shall be furnished with two diametrically opposite integral plastic ears for attaching the bail. The diameter of the wire used for the bail shall not be less than 7 gauge for class 1 and 9 gauge for class 2. Bail handle shall be a plastic material.

3.2.3 Covers (type I, class 1) Covers for type I, class 1 pails shall be steel. Covers may contain a closure with pouring spout when specified by the purchaser.

3.2.3.1 Thickness. Covers shall not be less than 0.0159 inches thick.

3.2.3.2 Types. Covers shall be the lug type or the closing ring type.

3.2.3.3 Seals. Covers shall accommodate a circular cross-section gasket or a flowed-in gasket seal.

3.2.3.4 Protective coatings.

3.2.3.4.1 Interior coatings. Unless otherwise specified, steel covers used to package materials having a deleterious effect on the covers (or vice versa), shall be given a smooth coat of an approved lining as specified by the pail purchaser.

3.2.3.4.2 Exterior coatings. All exterior surfaces of the steel covers shall be coated with rust-inhibiting enamel or equal. The coating shall have a minimum dry film thickness of 0.5 mil.

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3.2.4 Leakage (type I, class 1 and class 2). Covered filled pail shall not show any evidence of leakage when tested in accordance with 4.4.9.

3.2.5 Body (type I, class 2). The sides and bottom of the pail shall be an integral unit. The sides shall not be less than 0.065 inches thick and the bottom shall not be less than 0.080 inches thick. The top rim of the pail shall be curled down and may contain a reinforcing wire. The pail shall be tapered to nest to conserve space in storage and shipment. The interior surfaces of the pail shall be smooth and free of all cracks, crevasses, and recesses wherein product might be retained when pail is emptied. When fitted with covers, pails shall be designed so that the pails can be easily stacked.

3.2.6 Covers (type I, class 2). Covers for type I, class 2 shall be high density polyethylene. Covers may contain a closure with pouring spout when specified by the purchaser.

3.2.6.1 Thickness. Covers shall not be less than 0.070 inches thick.

3.2.6.2 Types. Cover shall be the friction snap on type or the closing ring type.

3.2.7 Body (type II). The pail body shall have a minimum thickness of 0.045 inches and shall be molded as an integral unit. The interior surfaces of the pail shall be reasonably smooth and free of all cracks, crevasses, and recesses wherein product might be retained when pail is emptied. Pails shall be sesigned so that the pails can be easily stacked. An example of such a design would be a recessed bottom and a raised top.

3.2.8 Closures (type II). Openings shall not exceed 2.7 inches. Closures shall be of a material resistant to the product to be contained and adequate to prevent leakage when tested in accordance with paragraph 4.4.8, 4.4.9, 4.4.10 and 4.4.11. Vented closures shall be provided when specified by the purchaser.

3.2.9 Handle (type II). The diameter of the wire used for the handle shall not be less than 5 gauge. Handle may also be plastic molded as an integral past of the pail. Handle shall be of such a design as not to interfere with the stocking of the pails.

3.2.10 Leakage (type II). Closed filled pail shall not show any evidence of leakage when tested in accordance with 4.4.8.

3.3 Capacity (volume). The capacity of the pail shall be rated at 5 gallons (U.S.), with a minimum capacity of 5.2 gallons (U.S.) (type I and II), a maximum capacity of 5.45 gallons (U.S) for type I and 5.75 gallons (U.S.) for type II.

3.4 Capacity (weight). Type I, class 1 pails shall be capable of holding up to 72 lbs. of product. Type I, class 2 pails shall be capable of holding up to 57 lbs. of product. Type II pails shall be capable of holding up to 47 lbs. of product.

3.5 Marking for identification.

3.5.1 Type I, class 1 pails shall be marked per National Motor Freight Classification package 2086 with the following information:

a. Manufacturer's name or trademark.
b. Minimum thickness.
c. Capacity in gallons.
d. Year of manufacture.
e. Letters NFC - to specify that the pails is non-returnable.
f. Grade (not a requirement of package 2086).

3.5.2 Type 1, class 2 pails shall be marked per National Motor Freight Classification package 2094 with the following information:

a. Manufacturer's name or trademark
b. Minimum thickness.
c. Capacity in gallons.
d. Year of manufacture.
e. Letters NFC - to specify that the pails is non-returnable.
f. Grade (not a requirement of package 2094).

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3.5.3 Type II pails shall be marked per DOT Specification 34 (49 CFR 178.19) with the following information:

(a) DOT - 34**; Stars to be replaced by the rated capacity of the container (for example DOT 34-5). These marks shall be understood to certify that the container complies with all specifications 34 requirements.

(b) Month and year of manufacture; name of maker or maker's symbol (symbol, if used, must be registered with the Bureau of Explosive). For example, "DOT-34-5-6/65 to indicate a container of 5 gallon capacity made in June 1965.

(c) Grade (not a requirement of specification 34).

3.6 Color. The color of the pails shall be specified by the purchaser (see 6.2c).

3.7 Leakage test. Closed filled pails shall not show any evidence of leakage when tested in accordance with 4.4.8.

3.8 Drop test. Closed filled pails shall not show any evidence of leakage when tested in accordance with 4.4.9

3.9 Drop test at low temperature. Closed filled pails shall not show any evidence of leakage when tested in accordance with paragraph 4.4.10.

3.10 Vibration test at elevated temperature. Closed filled pails shall not show any evidence of leakage when tested in accordance with paragraph 4.4.11.

3.11 Workmanship. The pails shall be free of fins, sharp edges and defects that may affect their durability or serviceability. The polyethylene shall be free from contamination (dirt and foreign material).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all pails submitted for inspection at the same time.

4.2.2 Visual examination of end item. Sampling for this examination shall according to level S-3 in MIL-STD-105. The Acceptable Quality Levels (AQL's) for defects listed in table I shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.2.3 Sampling for tests. Sampling for tests shall be in accordance with level S-2 in MIL-STD-105. The AQL for tests shall be 6.5 percent defective.

4.3 Examination.

4.3.1 End Item. Pails selected in accordance with 4.2.2 shall be examined for defects listed in table I.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Capacity not as specified	X	
Fins or sharp edges	X	
Markings illegible		X
Bail or handle does not have clearance or function properly	X	
Damage or defects not affecting function or serviceability		X
Damage or defects affecting function or serviceability	X	

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4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed in table II shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
Contents	Distortion of container. Incorrect count.

4.4 Tests.

4.4.1 General. Pails selected in accordance with 4.2.3 shall be tested to verify compliance with this specification.

4.4.2 Density. Density of the pail material shall be tested in accordance with ASTM D 1505 (68).

4.4.3 Tensile strength. Tensile strength of the pail material shall be tested in accordance with ASTM D 638 (68).

4.4.4 Elongation. Elongation of the pail material shall be tested in accordance with ASTM D 638 (68).

4.4.5 Stiffness. Stiffness of the pail material shall be tested in accordance with ASTM D 747 (70).

4.4.6 Carbon black. Percentage of carbon black in the pail material shall be determined in accordance with ASTM D 1603 (68).

4.4.7 Capacity test. The pail shall be filled with a measured amount of water to determine compliance with 3.3.

4.4.8 Leakage. The pail shall be filled to the top with water and allowed to stand for not less than one hour. The pail shall be fitted with a lid or closure then turned upside down and allowed to stand for not less than one hour. Examination shall then be made to determine compliance with 3.2.4 and 3.2.10.

4.4.9 Drop test. The pail shall be tested in accordance with method 5007 of Fed. Test Method Std. No. 101 using a 30 inch drop height and using water as a dummy load. Pail shall be dropped once flatwise on each end, plus one top edge drop and one bottom edge drop where center of gravity id directly above the striking point of the container. After the drop test the pail shall be inspected for leaks.

4.4.10 Drop test at low temperature. Pails must be filled to marked capacity with commodity and conditioned to 0 degrees F. or below for a minimum of 4 hours. Drop tests must be performed with the pail flat on its side and also at a 45 degree angle on the bottom chime onto solid concrete from a height of 48 inches. No container shall be required to withstand more than one drop. A minimum of three containers must be tested at each of the above mentioned areas without failure. (If commodity will not withstand the test temperature, a dummy load simulating the commodity as to weight and form may be used.)

4.4.11 Vibration test at elevated temperature. Three pails must be filled with commodity to marked capacity, conditioned at 130 degrees F. for 4 hours, stacked 3-hi and vibrated for one hour at 1 "G" to a vertical linear motion. Each pail must perfo without failure. (If commodity will not withstand the test temperature, a dummy load simulating the commodity as to weight and form may be used.)

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Each pail shall be packaged method III in accordance with MIL-P-116.

5.1.2 Level B. Each pail shall be packaged in accordance with 5.1.1.

5.1.3 Level C. The pails shall be packaged in a manner to assure protection from hazards under known favorable conditions during shipment, handling, and limited tenure of storage.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Pails shall be nested or stacked and then packed in boxes conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, overseas type, PPP-B-621, class 2, PPP-B-636, class weather-resistant, or PPP-B-640, class 2. Fiberboard meeting the requirements of PPP-F-320 shall be utilized for partitions between the nested or stacked pails. The fiberboard boxes shall be closed in accordance with the appendix to the box specification. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification. Strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Pails shall be nested or stacked and then packed in a close-fitting box conforming to PPP-B-636, class domestic, or class weather-resistant when specified. Fiberboard meeting the requirements of PPP-F-320, shall be utilized for partitions between nested or stacked pails. The box shall be closed in accordance with the appendix to the box specification. The gross weight shall not exceed the weight limitations of the box specification.

5.2.3 Level C. The pails shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the National Motor Freight Classification or Uniform Freight Classification rules or rules of other modes of transportation as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. General purpose grade plastic pails covered by this specification are intended to be used for shipping and storing products such as paint, and other liquids or dry materials. These pails are not be used for the transportation or storage of food or beverages.

Food and beverage grade plastic pails covered by this specification are intended for use in the transportation and storage of food and beverages.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of specification.
(b) Type, class and grade (see 1.2).
(c) Color.
(d) Size of lot, if different from 4.2.1.
(e) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

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